An Introduction to Agent-Based Modeling

Unit 3: Extending and Exploring Models

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Four Characteristics of ABM

1. Simple rules generate complex phenomena
2. Randomness at the level of the individual can result in nearly deterministic population-level behavior
3. Complex patterns can “self-organize” without any central leader
4. Different models emphasize different aspects of the world

The Problem (Arthur, 1994, AER)

- 100 people in Santa Fe, NM like Irish Music
- On Thursday an Irish band plays at El Farol (The Lantern)
- However if more than 60 people are at the bar then its crowded and no one has fun
- The newspaper publishes the attendance each week
- On the basis of the past *n* weeks how do individuals decide whether or not to go to the bar?

Arthur’s Experiment

- W. Brian Arthur postulated this problem and proposed a solution:
  - Individual have a bag of strategies to predict attendance such as:
    - Last week’s attendance times two
    - Two weeks ago’s attendance minus last week’s
  - Each agent determines which strategy would have worked the best had they used it the last five weeks
  - They then use that strategy to predict what this week’s attendance will be and decide whether to go to the bar on that basis

The Question of the El Farol Bar Problem

- Most humans do not act rationally, they act in a boundedly rational way
- Most humans do not reason deductively from first principles, they reason inductively from past experience
- If you assume agents participating in this way act this way, what are the results?

The NetLogo Version

- Each agent has:
  - a memory of *t* weeks
  - a set of *n* strategies of the form:
    - $a_t x_t + a_{t-1} x_{t-1} + \ldots + a_0$
    - So they strategy is defined by the a's
- The agent determines the total error between predicted and actual attendance of each of their strategies given their memory
- The agent utilizes the least erring strategy this timestep to predict the attendance (<=60 they go, and >60 they do not)

What is Success?

- The original model illustrates how many agents make it to the bar and whether or not the bar is crowded
- But what if we want to know which agents are doing better at attending the bar than other agents?

Action is the foundational key of all success. - Pablo Picasso

Adding Reward

- Modify turtles-own to add *reward*
- Initialize *reward* in turtle creation
- Make sure turtles get rewarded for going to the bar when it is not crowded
- Change the turtle’s color based on reward

Colors / Value in NetLogo

- Color Swatches
- Relationship between color names and numbers
- Value is a powerful way to represent ordinal data, and even gives you some insight into quantitative data

The Power of Visualization

- Your insight into your model is useless if you can’t convince others
- Not everyone is a modeler
- Visual tools are some of the best ways to reach people

Statistical Summaries

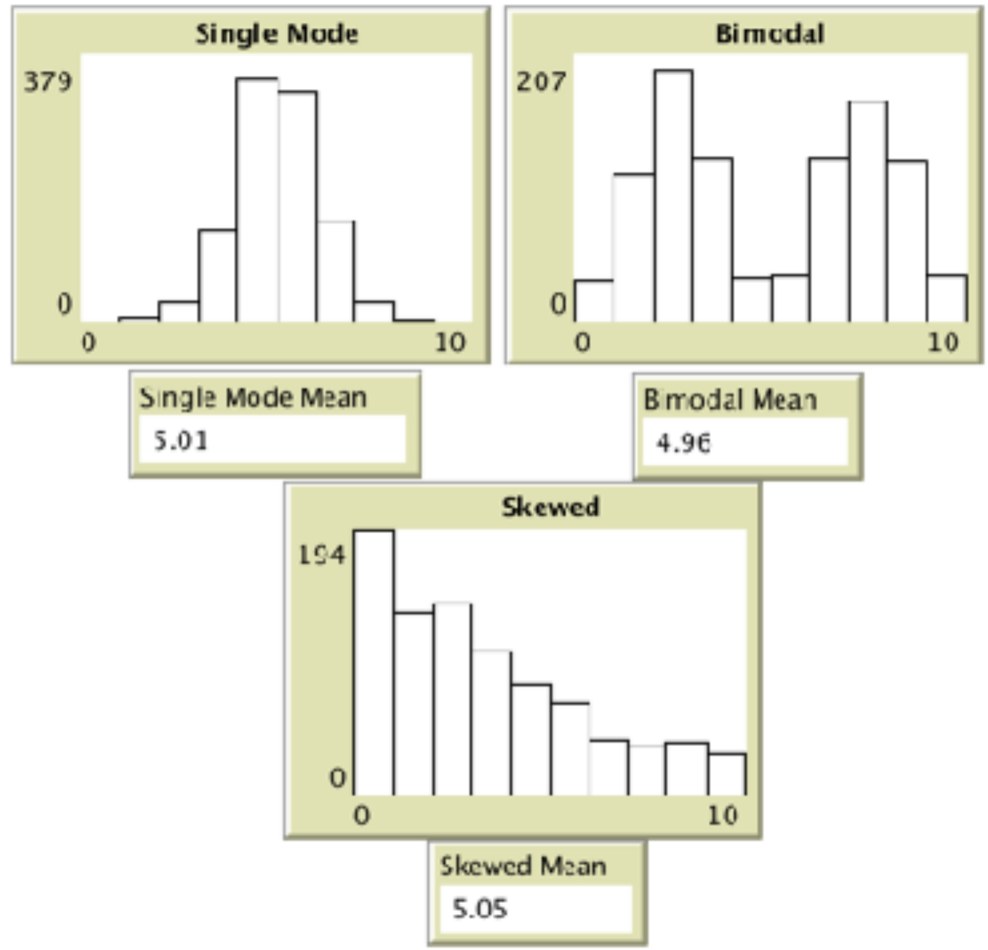
- Max reward
- Min reward
- Mean reward
- Median reward
    - Bonus: Snap to grid

Plotting

- Recreating the attendance plot
  - plot the attendance
- plot-pen-reset
- plotxy
  - plot 0 threshold
  - plot max threshold (plot-x-max)

Looking Beyond the Average

- Examining the whole distribution is often useful in agent-based modeling
- Bimodal distributions and skewed distributions are quite common and are not well described by means



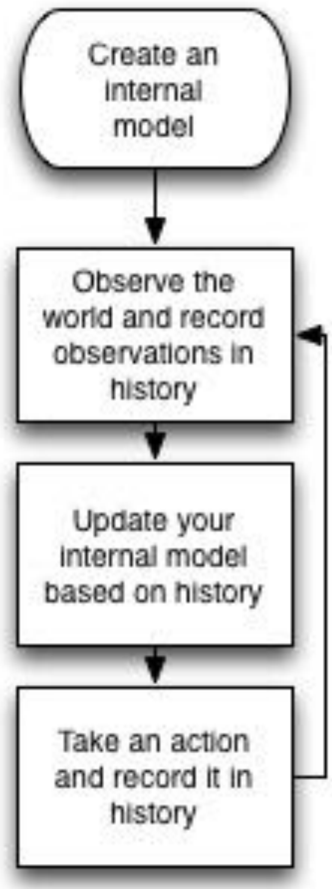
Histogram

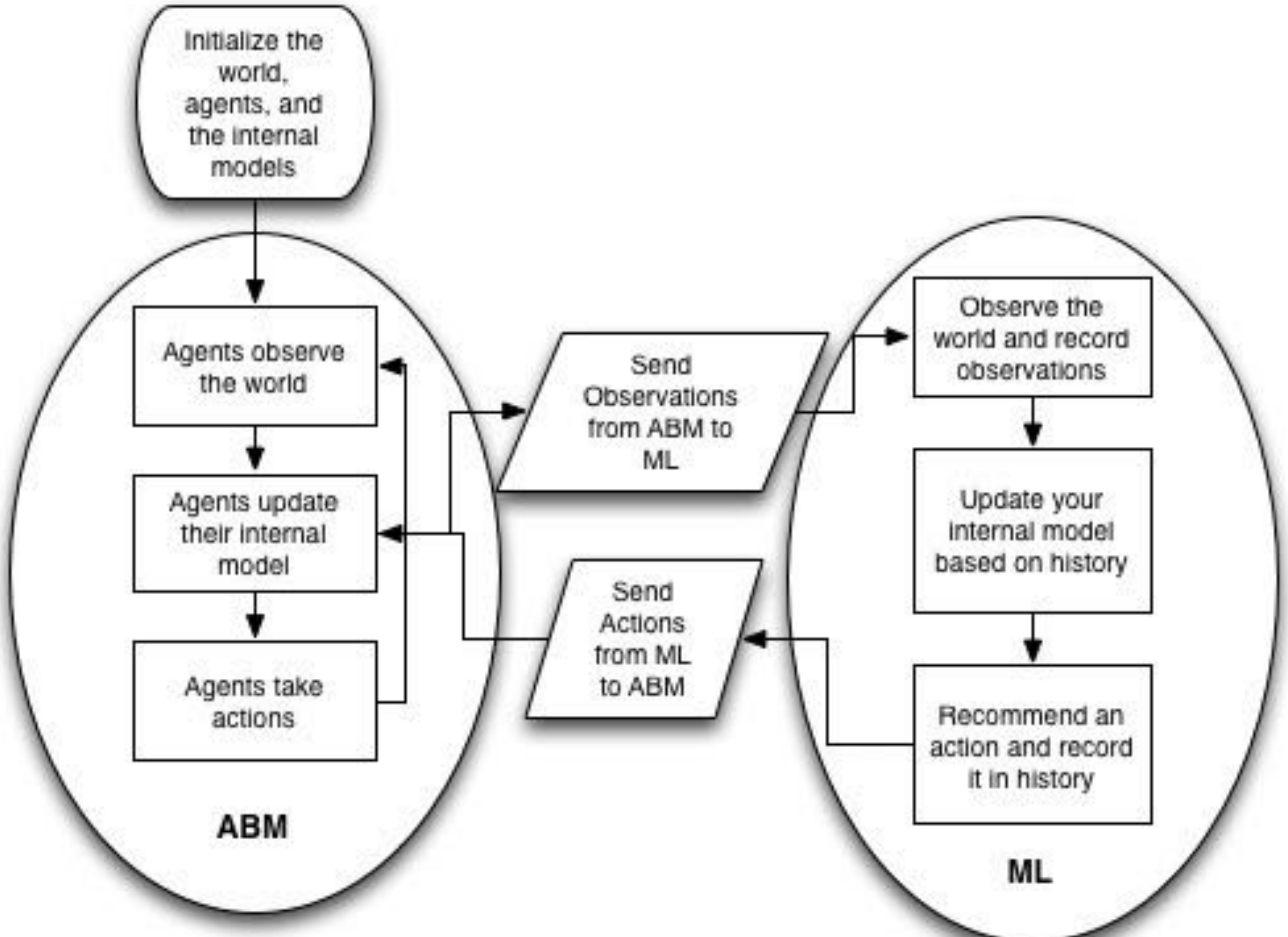
- Create a Histogram using the new plot
- set-plot-y-range
- set-plot-x-range

Machine Learning and ABM

The ABM Processing Loop

The ML Processing Loop





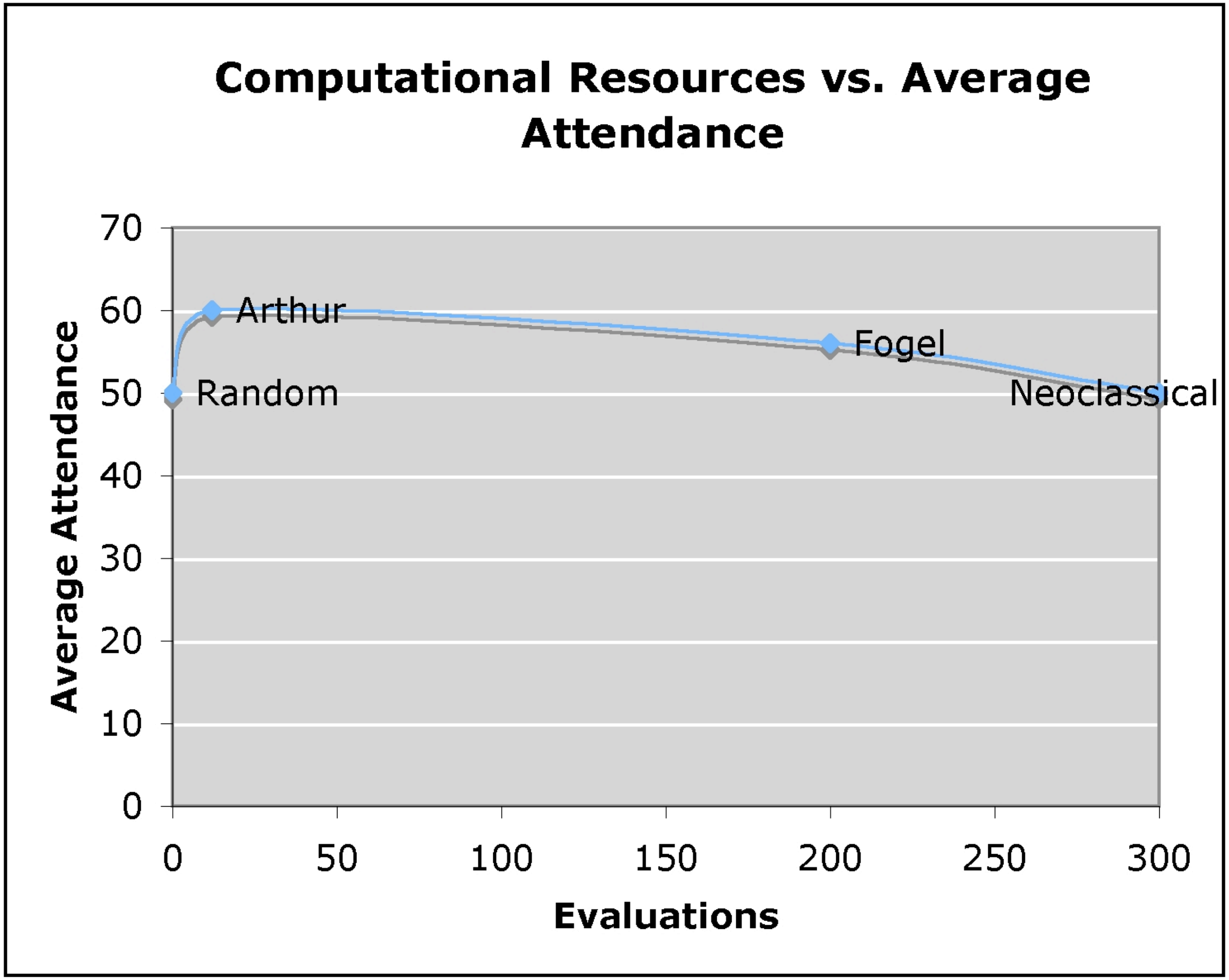
Initialize the world, agents, and the internal models
Agents observe the world
Agents update their internal model
Agents take actions
ABM
Send Observations from ABM to ML
Send Actions from ML to ABM
Observe the world and record observations
Update your internal model based on history
Recommend an action and record it in history
ML

Revisiting the El Farol Bar Problem

- At the end of Arthur's paper he says that if a GA was used instead of a bag of strategies the result would be the same
- Fogel et al. (1999) took up Arthur's challenge and used a GA
- He showed that the average attendance was 57
- He allowed each individual to run a GA for 25 generations with 100 different strategies

Observations

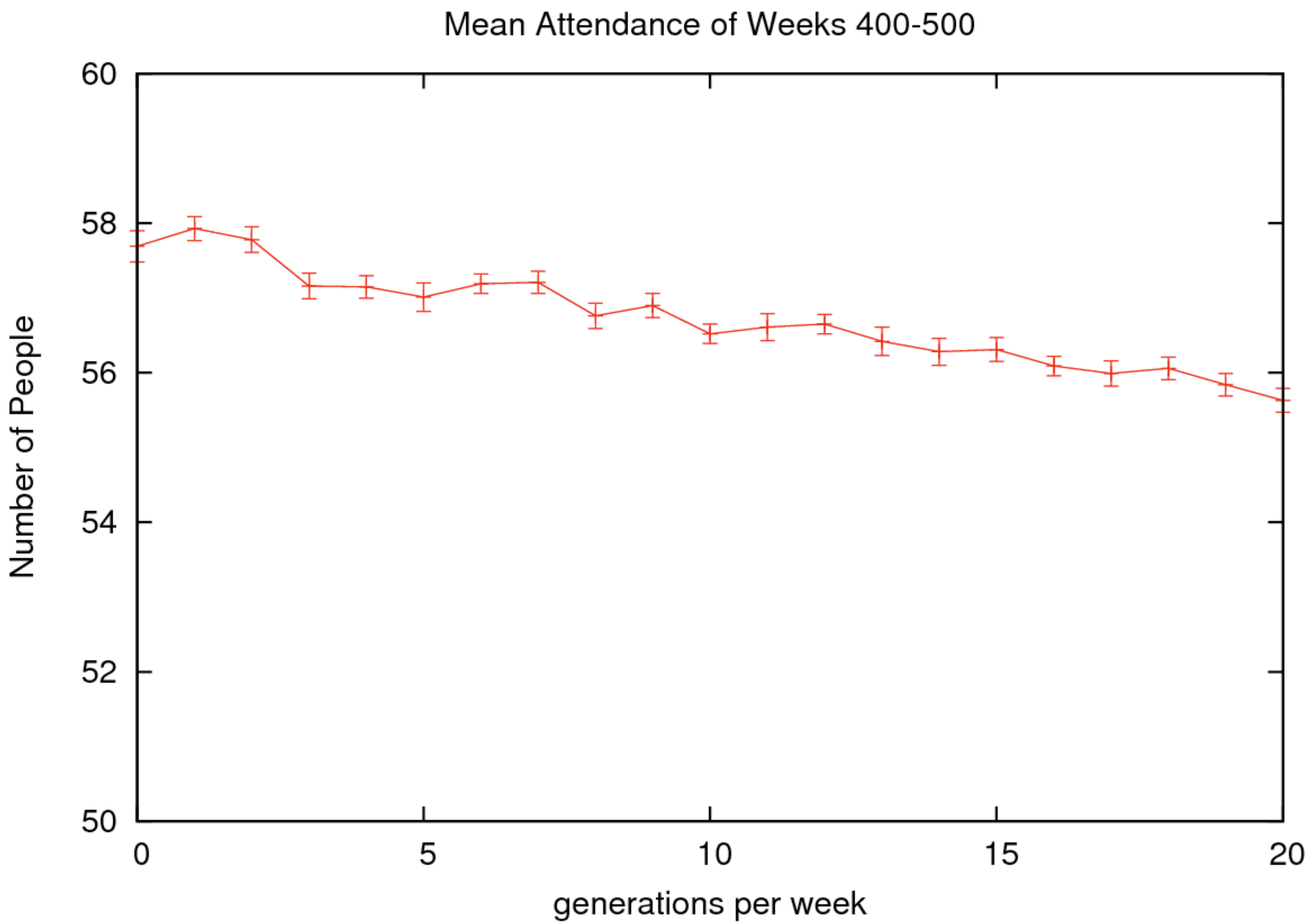
- Random Attendance is 50
- Arthur’s Attendance is 60
- Fogel et al. Achieves an Attendance of 57
- Perfectly Rational Agents - 50?
  - Normal Distribution over Predictions?
  - All Attend or All Don’t Attend each week?


Computational Resources vs. Average Attendance
Average Attendance
Evaluations
70
60
50
40
30
20
10
0
0
50
100
150
200
250
300
Arthur
Fogel
Random
Neoclassical

The Experiment

- Use a GA to optimize the strategy space
- Vary Number of Generations from 0 to 20
- Run the Model for 500 ticks
- Average over 30 runs
- Measure Mean Attendance per Week for last 100 weeks

In collaboration with Forrest Stonedahl



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Other NetLogo Models Related to El Farol

- Minority Game
  - Challet and Zhang, 1997
- Santa Fe Artificial Stock Market
  - Palmer, Arthur, Holland, and LeBaron, 1999
- Root Beer Game
  - Sterman, 1989

Unit 3 Overview

- El Farol Bar Problem
- Visualization
- Plotting
- Distributions
- Machine Learning + ABM
- Unit 3 Slides
- Brief Model Questionnaire
- Unit 3 Test